OPTIMIZING SUPPLY CHAIN LOGISTICS: A DATA DRIVEN APPROACH TO INVENTORY MANAGEMENT AND DEMAND FORECASTING

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Summer Training Program by IDRA

Data Science and AI

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India Data Research Academy

DECLARATION

I declare that this capstone report has been prepared for submission to the India Data Research Academy as part of the Data Science and AI Training Program. The report documents the analysis performed on the supply-chain dataset provided through the IDRA LMS. All external documentation and sources used to support the analytical workflow have been acknowledged in the references. The submitted notebook should be reviewed and understood by the student before final academic submission.

ACKNOWLEDGEMENT

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ABSTRACT

Effective inventory management depends on anticipating demand while balancing stock availability, replenishment decisions and operational constraints. This capstone investigates the supplied IDRA supply-chain dataset containing 91,250 daily records across 50 SKUs, 5 warehouses, 10 suppliers and 4 regions during 2024. The analysis targets Units Sold as a daily demand proxy and follows a complete data-science workflow covering data quality assessment, preprocessing, exploratory analysis, statistical analysis, feature engineering and regression modelling. The raw dataset contained no missing values or duplicate rows; date fields were standardized and potential outliers were screened using the IQR rule and retained when operationally plausible. Demand showed pronounced month-to-month seasonality, while promotion records were associated with a descriptive 27.74% higher average daily demand than non-promotion records. The supplied Demand Forecast was strongly correlated with actual demand ($r = 0.9493$) and achieved a test MAE of 2.37 units, providing a strong benchmark. A HistGradientBoostingRegressor using the existing forecast, lagged demand, rolling statistics, promotion and seasonal features reduced test MAE to 2.19 units and RMSE to 2.75, with R^2 of 0.793. The results support using the existing forecast as a strong planning signal while applying ML calibration, promotion-aware planning and time-sensitive inventory monitoring to improve operational decisions.

Keywords: Supply Chain Analytics; Demand Forecasting; Inventory Management; Exploratory Data Analysis; Regression; Predictive Analytics

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1. INTRODUCTION

1.1 Background

Supply chain operations require firms to maintain enough inventory to meet customer demand without tying up excessive working capital in stock. Demand varies over time because of seasonality, promotions and operational conditions, making static replenishment policies difficult to optimize. Data driven forecasting provides a way to combine historical demand with available operational signals so that inventory and purchasing decisions can be aligned more closely with expected demand.

1.2 Context of the Problem

The supplied dataset represents daily supply chain activity at SKU and warehouse level. It includes sales volume, inventory position, supplier lead time, reorder point, order quantity, pricing, promotion status and an existing demand forecast. These variables reflect the information that inventory planners and supply chain managers may use when deciding how much stock to hold and when to replenish. The analysis therefore focuses on two connected questions: how demand behaves across time and operating conditions, and whether a machine learning model can improve the supplied forecast as a demand-planning signal.

1.3 Scope of the Study

The study covers data quality assessment, descriptive and exploratory analysis, statistical summaries, demand related feature engineering, time aware regression modelling and practical supply chain recommendations. The prediction task is daily one step ahead estimation of Units Sold for the final portion of the observed calendar period. The study does not attempt to optimize a full multi-echelon inventory policy, calculate an economic order quantity from cost parameters, establish causal effects, or deploy the model into a live enterprise system.

2. PROBLEM STATEMENT AND MOTIVATION

The central problem is: How can historical demand, inventory and operational information be used to improve daily demand prediction and support more informed inventory-management decisions? Accurate demand estimates can reduce the risk of overstocking and understocking, help planners time replenishment, and make promotional periods easier to prepare for. The motivation for using machine learning is not to replace the supplied forecast blindly, but to test whether a nonlinear model can extract additional predictive signal from operational and temporal variables.

3. OBJECTIVES

1. Assess the structure and quality of the supplied supply chain dataset and prepare it for reliable analysis.
2. Identify demand patterns across time, promotion status, regions, warehouses and operational variables.
3. Quantify the relationships between actual demand, the supplied demand forecast, inventory and other numerical variables.
4. Engineer leakage-aware lag, rolling and seasonal features for daily demand prediction.
5. Build and evaluate regression models using a time-based train-test design and compare them with simple and existing-forecast baselines.
6. Translate the evidence into practical recommendations for demand planning and inventory management.

4. RESEARCH QUESTIONS

7. What are the main characteristics and data quality issues in the supplied supply-chain dataset?
8. How does demand vary over time and across promotion and operational conditions?
9. Which variables are most strongly associated with actual Units Sold?
10. How accurately can a machine-learning model predict daily demand on unseen future dates?
11. Does the final model improve on the existing Demand Forecast and simple lag-based baselines?
12. What inventory management actions are supported by the observed demand and reorder point patterns?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset used in this project is the project-specific CSV supplied through the India Data Research Academy LMS for Project 3, “Optimizing Supply Chain Logistics: A Data Driven Approach to Inventory Management and Demand Forecasting.” The capstone guide states that each approved project has a corresponding dataset available in the LMS and expects the assigned dataset to be used. The dataset was accessed for this analysis in September 2026. No external dataset was substituted.

5.2 Dataset Size and Structure

The raw dataset contains 91,250 observations and 15 variables covering 1 January 2024 through 30 December 2024. The records represent 50 SKUs observed across 5 warehouses, with 10 suppliers and 4 regions. Units Sold is selected as the prediction target because it represents realized daily demand/sales. Demand Forecast is treated as an existing planning forecast and used as a predictor under the assumption that it is available before the corresponding actual sales outcome.

Dataset characteristic	Value	Dataset characteristic	Value
Observations	91,250	Variables	15
Time period	01 Jan–30 Dec 2024	Unique dates	365
SKUs	50	Warehouses	5
Suppliers	10	Regions	4
Target	Units_Sold	Task	Regression / demand prediction

Table 1. Summary of the Dataset

5.3 Data Dictionary

Variable	Meaning	Data type	Role
Date	Record date	datetime (after cleaning)	Feature / time index
SKU_ID	Product/SKU identifier	Categorical	Feature / identifier
Warehouse_ID	Warehouse identifier	Categorical	Feature / identifier
Supplier_ID	Supplier identifier	Categorical	Operational feature
Region	Geographic region	categorical	Segmentation feature
Units_Sold	Units sold on the date	integer	Target
Inventory_Level	Inventory available	integer	Operational feature
Supplier_Lead_Time_Days	Supplier lead time	integer	Operational feature
Reorder_Point	Inventory reorder threshold	integer	Operational feature
Order_Quantity	Quantity ordered	integer	Operational feature
Unit_Cost	Unit procurement cost	numeric	Operational feature
Unit_Price	Unit selling price	numeric	Commercial feature
Promotion_Flag	Whether promotion was active	binary integer	Demand feature
Stockout_Flag	Stockout indicator	binary integer	Diagnostic field; constant zero
Demand_Forecast	Existing demand forecast	numeric	Planning signal / feature

Table 2. Data Dictionary

6. DATA CLEANING

The cleaning stage followed the guide's requirement to document missing values, duplicates, data types, inconsistent values and relevant outliers rather than cleaning blindly. The raw file was preserved conceptually as the source, while the cleaned dataset standardized the date field and added documented diagnostic fields.

6.1 Missing Values

No missing values were present in any of the 15 raw columns. Therefore, no imputation was required and no observations were dropped for missingness. This preserves the full 91,250-row sample.

6.2 Duplicate Records

No exact duplicate rows were detected. The dataset therefore remained at 91,250 observations after duplicate validation.

6.3 Data Types and Inconsistent Values

The Date column was initially stored as an object/string and was converted to datetime for temporal analysis and feature engineering. Categorical identifiers were inspected and contained consistent structured labels for 50 SKUs, 5 warehouses, 10 suppliers and 4 regions. No inconsistent capitalization or unexpected category labels were identified.

6.4 Outlier Handling

Numerical variables were screened using the interquartile-range (IQR) rule. The screening flagged 110 Units Sold records, 199 Inventory Level records and 185 Demand Forecast records. Order Quantity was zero for 94.49% of records, making a conventional IQR fence unsuitable because Q1 and Q3 were both zero; the non-zero order quantities were therefore treated as a zero-inflated operational pattern rather than automatically removed. The flagged values were retained because the observed ranges are plausible in a supply-chain context and extreme demand/inventory observations can themselves be operationally informative. No values were removed solely because they were statistically unusual.

6.5 Before vs After Cleaning

Quality check	Before cleaning	Action	After cleaning
Rows	91,250	No row deletion required	91,250
Columns	15	Retained source variables	15 source variables
Missing cells	0	No imputation needed	0
Duplicate rows	0	No deletion needed	0
Date type	Object/string	Converted with pd.to_datetime	Datetime
Category labels	Structured labels	Inspected; no corrections required	Consistent
Outliers	IQR flags present	Reviewed and retained when plausible	No unjustified deletion
Stockout flag	One unique value; all 0	Retained for documentation; excluded from model	All 0

Table 3. Data Quality Before and After Cleaning

7. DATA PREPROCESSING

Preprocessing was designed around the regression objective and the temporal nature of the data. The main transformations were feature creation and time-aware splitting. Because the final HistGradientBoostingRegressor uses numerical inputs, no one-hot encoding was required for the final model. Identifier columns were used for EDA/segmentation but not included as raw predictive inputs.

7.1 Categorical Encoding

SKU_ID, Warehouse_ID, Supplier_ID and Region are categorical identifiers. They were not one-hot encoded in the final model because the forecasting formulation focuses on measurable demand, commercial and operational signals; using raw identifiers would increase dimensionality without providing a natural numeric meaning. The identifiers remain available for subgroup analysis and future hierarchical forecasting.

7.2 Feature Scaling and Transformation

HistGradientBoostingRegressor does not require standard scaling for numerical features, so scaling was not applied. Instead, the analysis created lagged demand, rolling statistics and cyclical time features. All rolling statistics were shifted by one day before calculation so that the current target could not contribute to its own predictors.

7.3 Train-Test Split

A chronological 80:20 split was used. The first 292 calendar dates (through 18 October 2024) form the training period and the final 73 dates (19 October through 30 December 2024) form the test period. This is preferable to random shuffling for a forecasting problem because it evaluates performance on later dates and avoids using future observations to predict earlier ones.

7.4 Final Feature Set

The final input set contains inventory and supplier variables, price variables, promotion status, lagged demand, rolling demand statistics, calendar features, cyclical seasonal features, a price-cost ratio and the existing Demand Forecast. The target is Units_Sold. Stockout_Flag is excluded. The Demand_Forecast feature is subject to the stated planning-time availability assumption.

Final predictors: Inventory_Level, Supplier_Lead_Time_Days, Reorder_Point, Order_Quantity, Unit_Cost, Unit_Price, Promotion_Flag, Lag_1, Lag_7, Rolling_Mean_7, Rolling_Mean_30, Rolling_Std_7, DayOfWeek, Month, DayOfMonth, IsWeekend, Price_Cost_Ratio, Demand_Forecast, DayOfYear, Month_Sin, Month_Cos, DayOfYear_Sin, DayOfYear_Cos.

8. EXPLORATORY DATA ANALYSIS

8.1 Univariate Analysis

The distribution of Units Sold is centered around 20 units per record, with mean 20.05 and median 20.00. The standard deviation is 9.07 units and the IQR is 14 units. The histogram shows substantial spread rather than a single fixed demand level, supporting the need for temporal and contextual predictors.

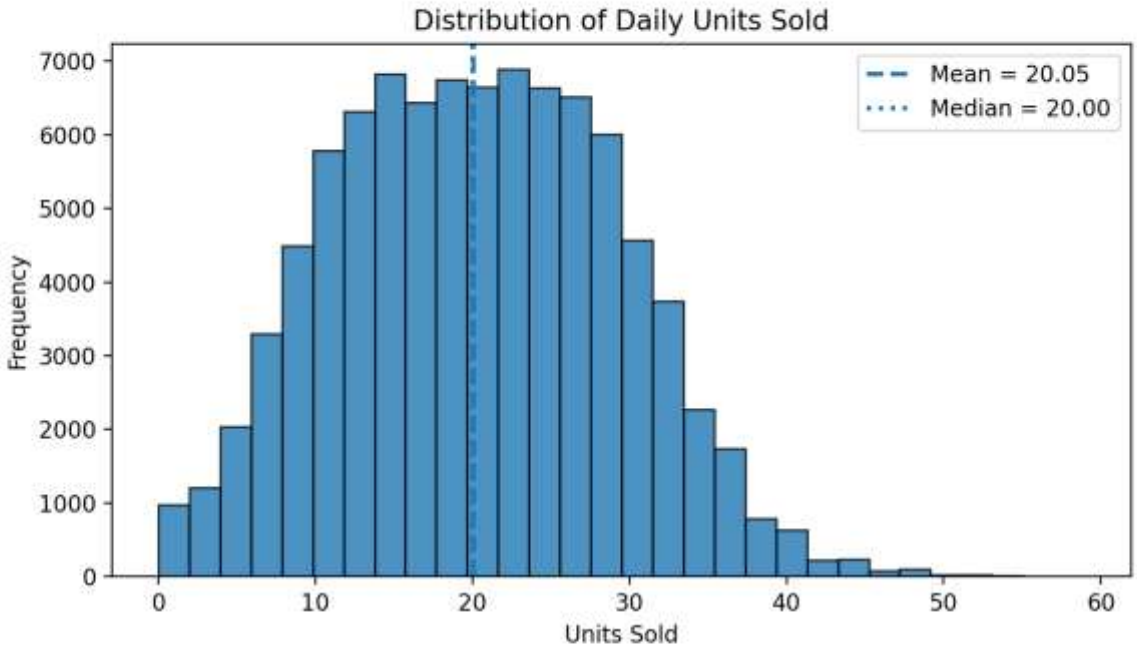

Figure 1. Distribution of Daily Units Sold

8.2 Categorical Analysis

Promotion status provides a clear descriptive difference in demand. Records with a promotion have mean Units Sold of 24.91 compared with 19.50 without a promotion, a difference of approximately 27.74%. This is an association in the supplied data rather than proof that promotions causally generate the full increase, because other conditions may differ between promotional and non-promotional records.

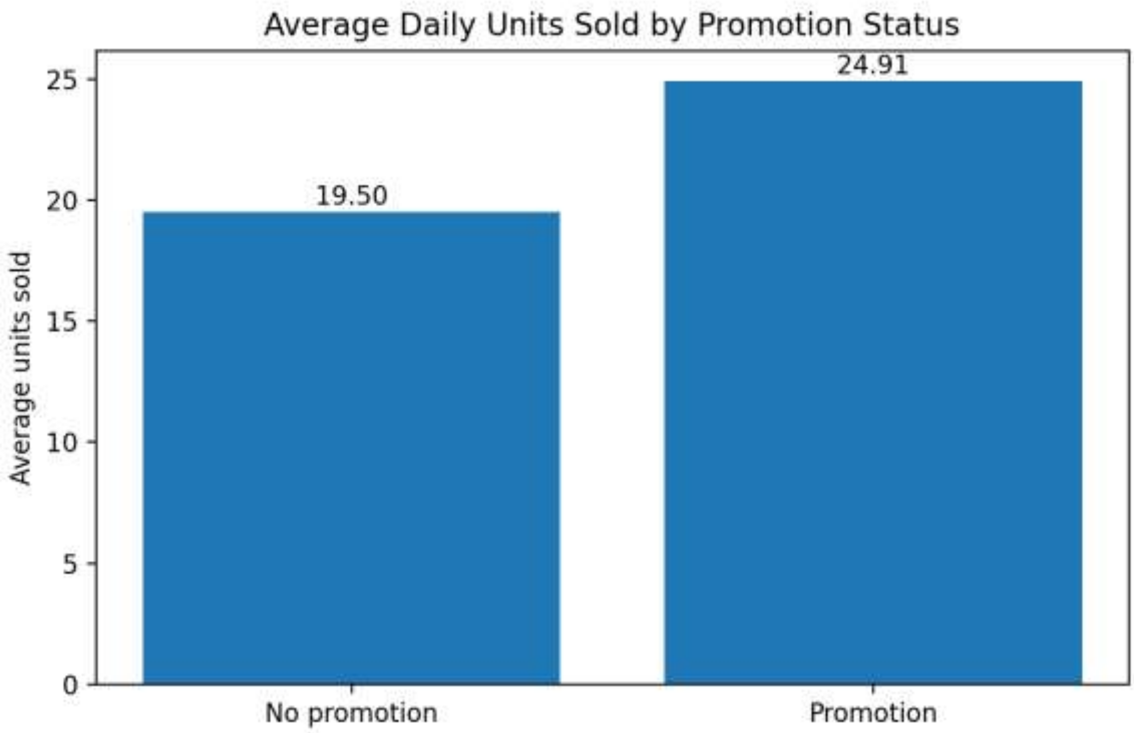

Figure 2. Average Daily Units Sold by Promotion Status

8.3 Bivariate / Relationship Analysis

The monthly demand pattern shows pronounced seasonality. Average demand rises from 22.80 units in January to 29.88 in March, then declines to 10.17 in September before recovering to 17.59 in December. The existing Demand Forecast closely tracks the same seasonal pattern. This supports including calendar and cyclical features in the forecasting model.

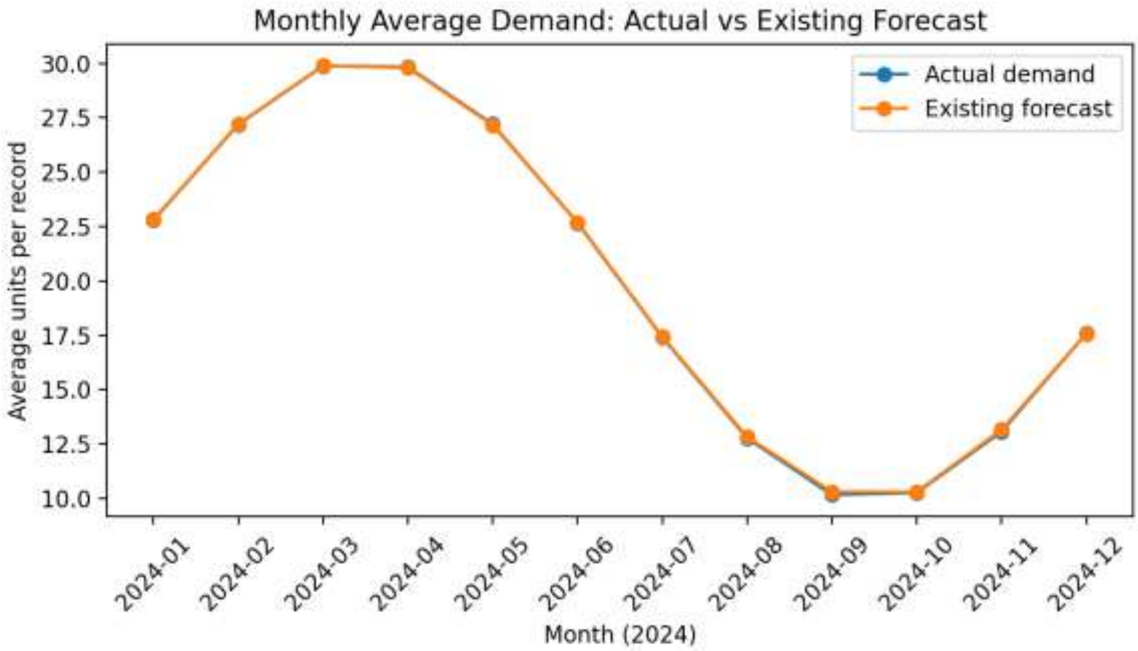

Figure 3. Monthly Average Demand: Actual vs Existing Forecast

8.4 Correlation Analysis

Demand Forecast has the strongest numerical association with Units Sold, with Pearson correlation r = 0.9493. Promotion Flag has a smaller positive association (r = 0.1802), while Inventory Level has a weak negative association (r = −0.0643). The relationships indicate that the supplied forecast contains substantial predictive information, while promotion provides additional demand signal. Correlation is used here for association and feature assessment, not causal inference.

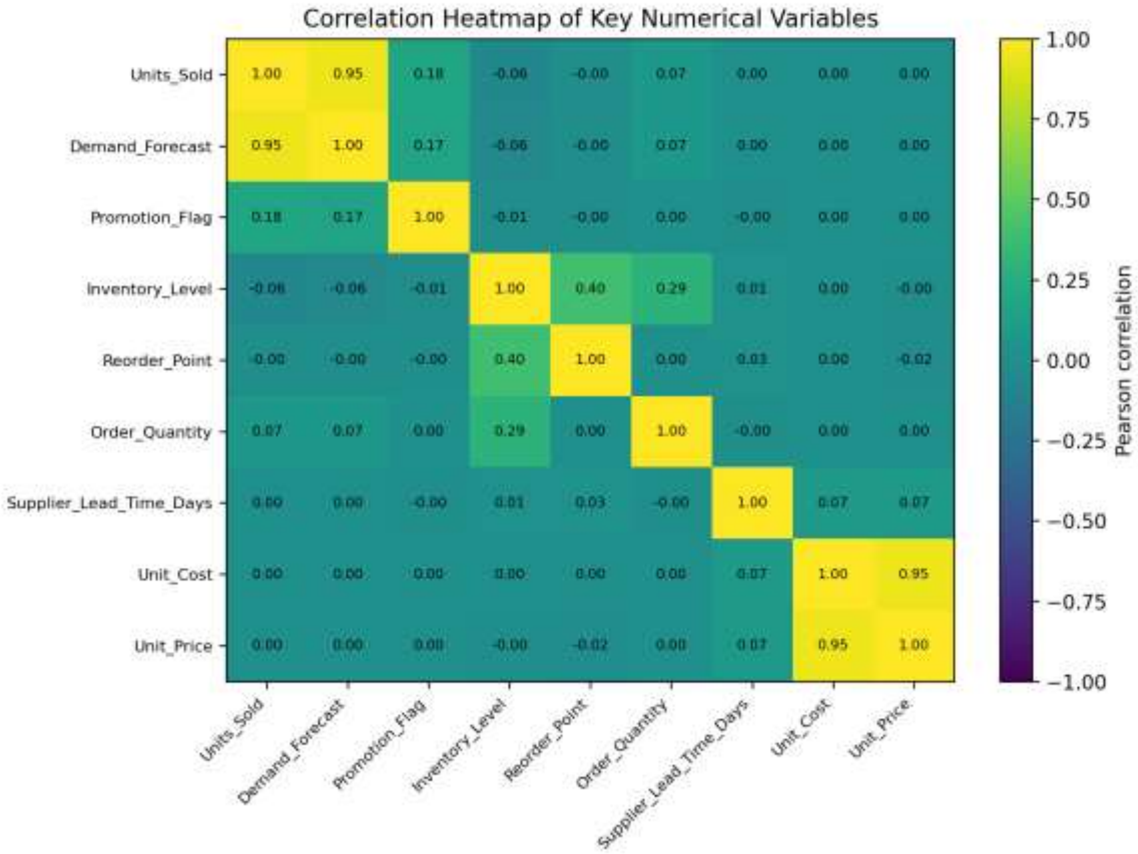

Figure 4. Correlation Heatmap of Key Numerical Variables

8.5 Other Figures

Inventory management was assessed using the share of records whose Inventory Level was below the stated Reorder Point. Across the full dataset, 5.25% of records fall below this threshold. The rate is higher during the higher-demand months of February through May and lower around September–October, suggesting that replenishment monitoring should be more attentive during periods of elevated demand.

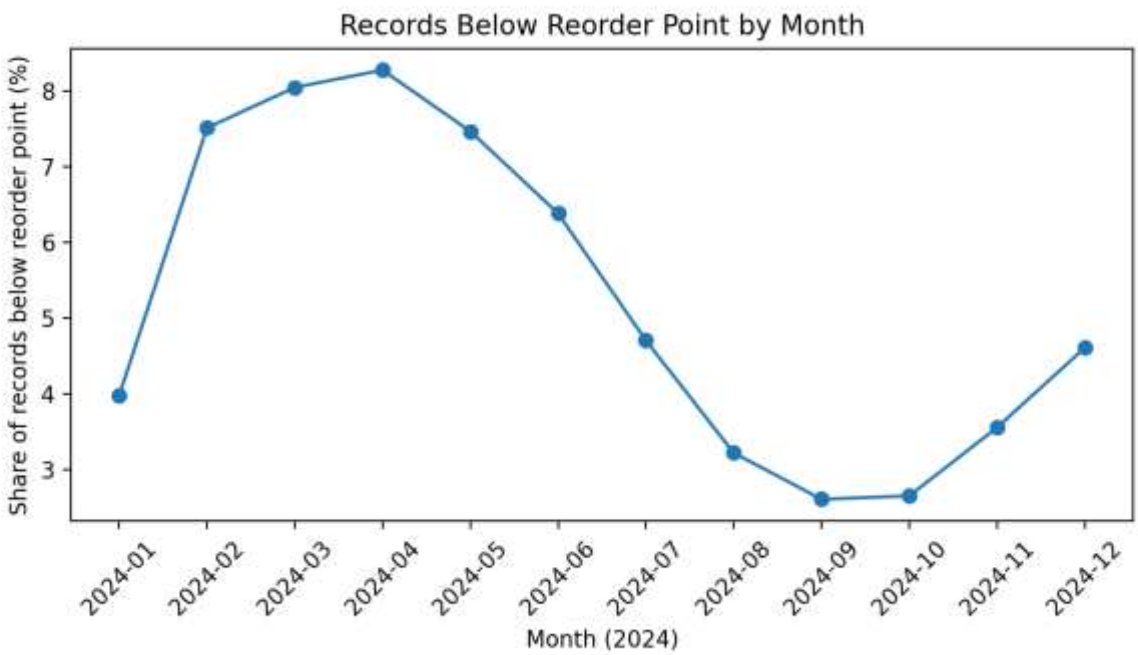

Figure 5. Records Below Reorder Point by Month

9. STATISTICAL ANALYSIS

Descriptive statistics were used to characterize central tendency, dispersion and operational variability. Table 4 shows that Units Sold and Demand Forecast have very similar centers, while inventory and order quantities show substantially different distributions. The median Order Quantity is zero because most records have no order placed on that date.

Variable	Mean	Median	Mode	Std Dev	Variance	IQR
Units_Sold	20.055	20.000	20.000	9.069	82.240	14.000
Demand_Forecast	20.082	19.950	0.000	9.504	90.325	13.980
Inventory_Level	471.522	461.000	369.000	133.488	17819.047	194.000
Reorder_Point	300.068	300.000	218.000	54.880	3011.808	94.000
Order_Quantity	19.272	0.000	0.000	82.341	6780.012	0.000
Unit_Price	18.262	18.180	23.340	7.121	50.711	11.390

Table 4. Descriptive Statistics of Key Variables

The existing Demand Forecast has mean 20.08 versus actual mean 20.05, giving a mean forecast bias of only 0.027 units when defined as forecast minus actual. Its correlation with actual demand is 0.9493, confirming that it is already a strong planning signal. The promotion comparison indicates a sizeable descriptive demand difference, while the weak inventory-demand correlation suggests that higher inventory is not simply associated with higher realized demand in the supplied observations.

Statistical interpretation note: The reported correlations and group summaries describe relationships in the observed data; they do not establish causation. The capstone guide specifically recommends interpreting descriptive statistics carefully and avoiding causal claims from correlation alone.

10. FEATURE ENGINEERING

Feature engineering converts the daily transaction history into predictors that represent recent demand, longer-term demand level and seasonality. Lag features use only prior observations within the same SKU-warehouse series. Rolling features are shifted before aggregation so that the current day's target cannot leak into the predictors.

Engineered feature	Construction	Purpose	Leakage control
Lag_1	Previous day Units_Sold	Captures short-term demand persistence	Uses only prior data
Lag_7	Units_Sold seven days earlier	Captures weekly recurrence	Uses only prior data
Rolling_Mean_7	Mean of previous 7 days	Smooths short-term demand noise	Shifted by one day
Rolling_Mean_30	Mean of previous 30 days	Represents local baseline demand	Shifted by one day
Rolling_Std_7	Std. dev. of previous 7 days	Captures recent demand volatility	Shifted by one day
DayOfWeek / Month / DayOfMonth	Calendar components	Captures calendar effects	Known before prediction
Month/Day-of-year sin/cos	Cyclical calendar encoding	Represents seasonal continuity	Known before prediction
Price_Cost_Ratio	Unit_Price / Unit_Cost	Adds a normalized commercial signal	Known operationally
Below_Reorder_Point	Inventory_Level < Reorder_Point	Supports inventory monitoring	Diagnostic; not model target

Table 5. Engineered Features

11. MODEL DEVELOPMENT

11.1 Problem Formulation

The machine-learning task is regression. The target variable is Units Sold, a continuous/count-valued measure of realized daily demand. The model predicts one day at a time for later dates using information available from the current operational record and historical demand. This formulation directly supports demand forecasting and downstream inventory planning.

11.2 Model Selection

Linear Regression was selected as an interpretable baseline. The final model is HistGradientBoostingRegressor, a tree-based boosting method capable of representing nonlinear relationships and interactions. This is appropriate because demand can respond differently under promotional periods, seasonal conditions and different recent-demand states. Simple lag-1 and lag-7 baselines were also evaluated, as was the existing Demand Forecast supplied with the dataset.

11.3 Training the Model

The final model used 300 boosting iterations, a learning rate of 0.05, a maximum of 31 leaf nodes per iteration, L2 regularization of 1.0 and random_state 42. The model was trained only on the first 80% of calendar dates and evaluated on the final 20%.

```
model = HistGradientBoostingRegressor(
    max_iter=300,
    learning_rate=0.05,
    max_leaf_nodes=31,
    l2_regularization=1.0,
    random_state=42
)
model.fit(train[features], train[target])
pred = model.predict(test[features])
```

The complete computational workflow remains in the accompanying Jupyter notebook. The report includes only the key training logic, consistent with the capstone guide.

12. MODEL EVALUATION

12.1 Regression Evaluation

The models were evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and R-squared (R^2). MAE represents the average absolute prediction error in units; MSE penalizes larger errors more strongly; RMSE expresses error in the original unit scale; and R^2 measures the proportion of target variance explained relative to a mean-based benchmark.

Model	MAE	MSE	RMSE	R^2
Existing Demand Forecast	2.366	8.831	2.972	0.759
Naive Lag-1	5.923	55.890	7.476	-0.525
Naive Lag-7	5.940	55.662	7.461	-0.519
Linear Regression	2.393	8.968	2.995	0.755
HistGradientBoosting (Final)	2.193	7.580	2.753	0.793

Table 6. Regression Model Performance

The final HistGradientBoosting model achieves test MAE 2.193 units, RMSE 2.753 units and R^2 0.793. It improves on the existing Demand Forecast baseline, whose test MAE is 2.366 and RMSE is 2.972. The corresponding reductions are approximately 6.31% in MAE and 6.33% in RMSE. The naive lag-1 and lag-7 baselines perform substantially worse, indicating that recent demand history alone is not sufficient for this dataset.

12.2 Model Performance Visualisation

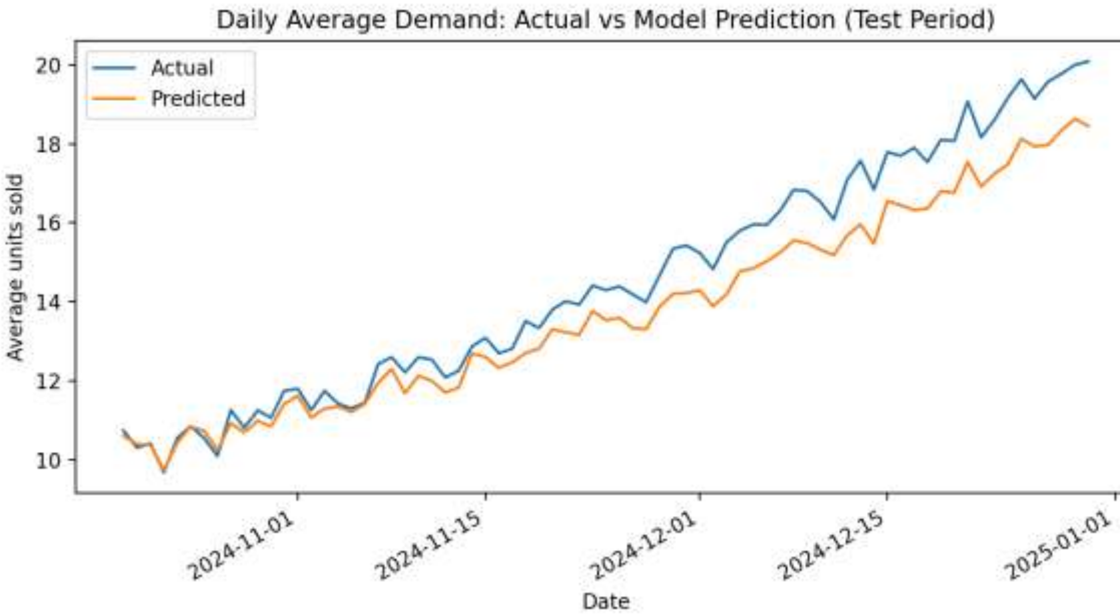

Figure 6. Daily Average Demand: Actual vs Model Prediction (Test Period)

The test-period time series shows that the model follows the broad movement in demand while smoothing some day-level variation. Because the final test period contains later seasonal conditions than the training period, some underprediction is visible toward the end of the horizon; this is reflected in the test error and is discussed as a modelling limitation.

12.3 Overfitting and Underfitting

The final model has training MAE 2.031 and test MAE 2.193; training RMSE is 2.548 versus test RMSE 2.753. Training R^2 is 0.928 compared with test R^2 0.793. The gap indicates some degree of fitting to the training data, but the relatively small increase in MAE from training to testing and the strong test R^2 suggest reasonable generalisation rather than severe overfitting. The chronological test design provides a more realistic check than a randomly shuffled split.

12.4 Prediction Error Distribution

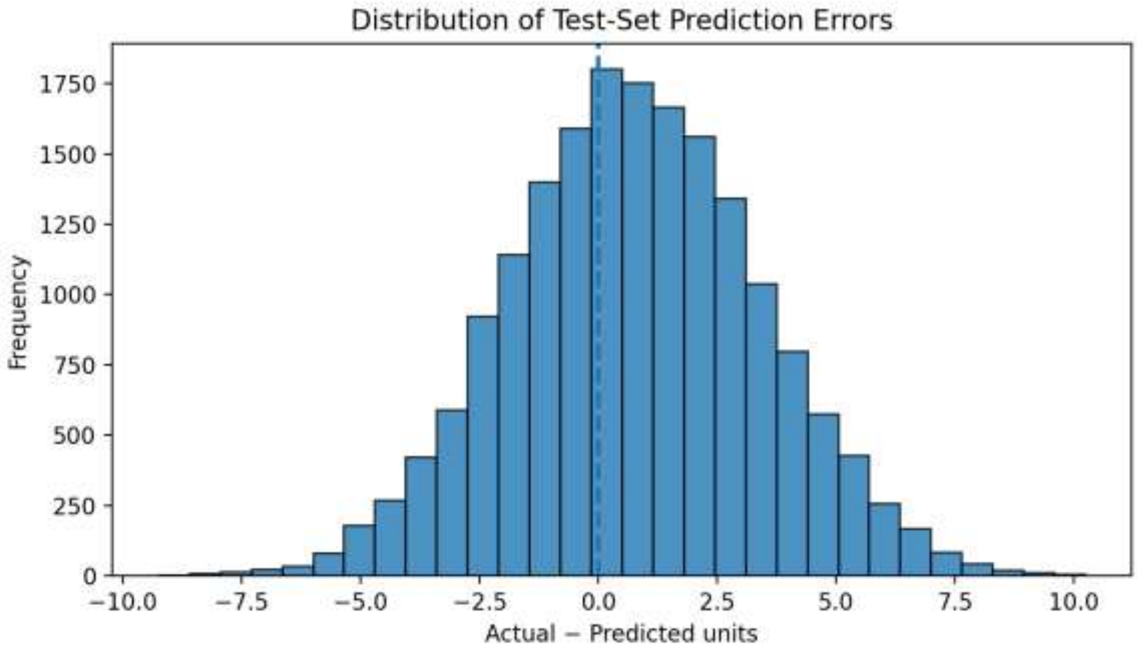

Figure 7. Distribution of Test-Set Prediction Errors

13. RESULTS AND FINDINGS

13. Finding 1 — The dataset is structurally clean. All 91,250 raw observations are usable without missing-value imputation or duplicate removal, and the date field can be standardized consistently.
14. Finding 2 — Demand is strongly seasonal. Average demand rises from 22.80 units in January to approximately 29.88 in March, then declines to about 10.17 in September before recovering to 17.59 in December. This supports explicit seasonal feature engineering.
15. Finding 3 — Promotions are associated with higher demand. Promotion records average 24.91 units versus 19.50 without promotion, a descriptive increase of approximately 27.74%. Promotion Flag is also among the most useful model inputs.
16. Finding 4 — The supplied Demand Forecast is already highly informative. It has correlation 0.9493 with actual Units Sold and a test MAE of 2.366 units, making it a strong operational benchmark rather than a weak starting point.
17. Finding 5 — ML calibration adds measurable improvement. HistGradientBoosting reduces test MAE to 2.193 units and RMSE to 2.753, improving on the supplied forecast by approximately 6.31% and 6.33%, respectively. Permutation importance identifies Demand Forecast as the dominant signal, followed by Promotion Flag, seasonal day-of-year structure and the 30-day rolling demand level.

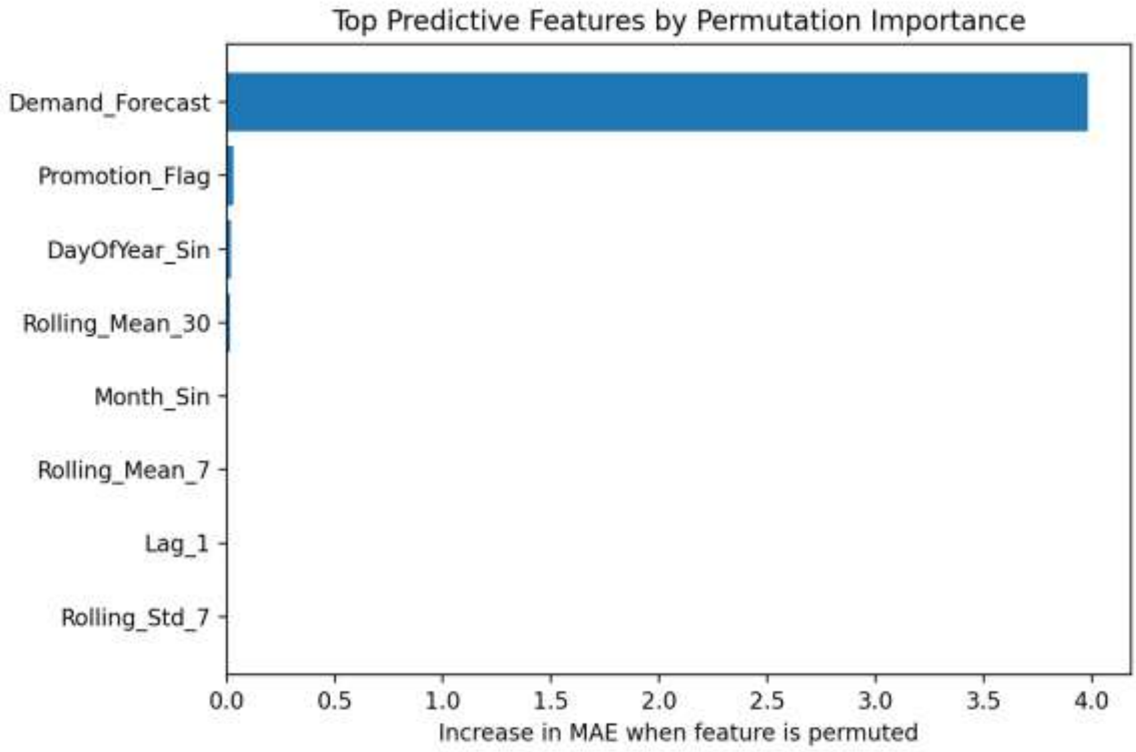

Figure 8. Top Predictive Features by Permutation Importance

14. DISCUSSION

The results indicate that the dataset already contains a strong demand-planning signal in Demand Forecast. Rather than replacing that signal, the machine-learning approach works best as a calibration layer that incorporates promotion status, seasonal structure and historical demand behaviour. This is operationally attractive because the model can be positioned as an enhancement to an existing forecasting process instead of requiring planners to discard established forecasts.

The seasonal pattern is especially important for inventory planning. Demand is materially higher in the first half of the year and materially lower around late summer and early autumn. A static inventory target would therefore risk holding excessive stock during lower-demand periods or becoming relatively less protective during higher-demand periods. The below-reorder analysis also shows that 5.25% of all records are below the defined reorder point, with higher rates during several high-demand months.

Promotion is a second meaningful signal. The descriptive demand difference suggests that promotion-aware replenishment can be useful, although the analysis cannot establish that promotion alone caused the increase. The model's permutation importance places Promotion Flag among the strongest non-forecast signals, supporting its inclusion in planning workflows.

From a model-governance perspective, the key assumption is the timing of Demand Forecast. The final model is valid as a forecast-calibration model only if this field is available before the actual Units Sold outcome. If the field was generated retrospectively using information from the same day or later, it would constitute leakage and should be excluded. The report therefore treats this as an explicit assumption and limitation rather than silently ignoring the issue.

15. LIMITATIONS

- The supplied dataset covers one calendar year. A longer history would provide stronger evidence for recurring annual seasonality and more robust validation across multiple cycles.

- The Stockout_Flag variable is constant at zero, so the study cannot empirically model stockout occurrence or quantify service-level risk from observed stockout events.
- Demand Forecast is a very strong predictor. The model's improvement is therefore incremental; it should be interpreted as forecast calibration rather than a completely independent forecasting system.
- The timing and construction of Demand_Forecast are not documented in the CSV. Its use assumes it was available before actual demand was realized; otherwise it could create target leakage.
- The analysis is predictive and descriptive. Correlations and promotion differences should not be interpreted as causal effects.
- The final validation is a single chronological holdout period. Rolling-origin cross-validation over multiple forecast windows would provide a stronger estimate of performance stability.

16. CONCLUSION

This capstone demonstrates a complete data-science workflow for supply-chain demand analysis and forecasting using the IDRA Project 3 dataset. The data were found to be clean and structurally consistent, with no missing values or duplicate rows. EDA and statistical analysis revealed strong seasonal variation in demand, a sizeable descriptive association between promotions and demand, and a very strong relationship between the supplied Demand Forecast and actual Units Sold.

A time-aware HistGradientBoostingRegressor provided the best tested predictive performance, achieving MAE 2.193, RMSE 2.753 and R^2 0.793 on the unseen final 20% of dates. Relative to the supplied Demand Forecast, the model reduced MAE and RMSE by approximately 6.31% and 6.33%. The overall conclusion is that the existing forecast is already a strong planning input, while a nonlinear ML calibration layer can provide measurable additional accuracy when combined with promotion, historical-demand and seasonal signals.

The project therefore supports a practical strategy of forecast-assisted inventory management: retain the strong existing forecast, calibrate it with leakage-controlled ML, and use seasonal and promotion-aware demand signals to inform replenishment monitoring.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

- Use the existing Demand Forecast as the baseline planning signal and apply the ML model as a calibration layer, because the existing forecast already has r = 0.9493 with actual demand while the ML model improves test MAE by about 6.31%.
- Increase inventory-planning attention during higher-demand months, particularly the February–May period, because average demand and the share of records below reorder point are both higher in these months.
- Make promotion schedules part of replenishment planning. Promotion records show approximately 27.74% higher average demand, and Promotion Flag is one of the strongest non-forecast predictive variables.
- Monitor inventory against reorder points using a time-sensitive dashboard. About 5.25% of all records are below the stated reorder point, so planners should review these cases before they become service risks even though the supplied stockout flag contains no positive events.
- Before operational deployment, verify that Demand Forecast is generated strictly before actual sales and establish automated retraining and forecast-error monitoring so model drift can be detected.

17.2 Future Work

- Collect multiple years of data and use rolling-origin time-series validation to test seasonal generalisation more rigorously.
- Add product category, customer/order-level, holiday, promotion type, price-discount depth and regional economic variables if available.
- Develop explicit stockout-risk and service-level models once reliable positive stockout observations are available.
- Compare the final model with dedicated time-series approaches and tuned ensemble models, while maintaining leakage-safe temporal validation.
- Translate demand forecasts into inventory policy metrics such as safety stock, reorder quantity and expected service level once reliable lead-time and demand-variability assumptions are available.

18. REFERENCES

India Data Research Academy. (2026). P_3_supply_chain_dataset1.csv [Dataset]. Data Science & AI Training Program, IDRA LMS.

India Data Research Academy. (2026). Capstone Project Guide 2026. Data Science & AI Training Program, IDRA.

India Data Research Academy. (2026). IDRA Capstone Project Template Sample. Data Science & AI Training Program, IDRA.

McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.

pandas development team. (2026). pandas documentation. Python Data Analysis Library.

scikit-learn developers. (2026). scikit-learn User Guide. Machine Learning in Python.

Matplotlib development team. (2026). Matplotlib documentation. Visualization with Python.

APPENDIX A. CODE

The complete notebook contains the full reproducible workflow. The following excerpts show the key analytical decisions without reproducing the entire notebook.

A.1 Data Loading and Inspection

```
df = pd.read_csv("P_3_supply_chain_dataset1.csv")
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
print(df.shape)
display(df.isna().sum())
print("Duplicate rows:", df.duplicated().sum())
```

A.2 Data Cleaning

```
# IQR screening
q1, q3 = df[col].quantile([0.25, 0.75])
iqr = q3 - q1
lower, upper = q1 - 1.5 * iqr, q3 + 1.5 * iqr
outliers = ((df[col] < lower) | (df[col] > upper)).sum()
```

A.3 Preprocessing and Feature Engineering

```
group = df.groupby(["SKU_ID", "Warehouse_ID"])["Units_Sold"]
df["Lag_1"] = group.shift(1)
df["Lag_7"] = group.shift(7)
df["Rolling_Mean_30"] = (
    df.groupby(["SKU_ID", "Warehouse_ID"])["Units_Sold"]
      .transform(lambda s: s.shift(1).rolling(30).mean())
)
df["DayOfYear_Sin"] = np.sin(2*np.pi*df["Date"].dt.dayofyear/365)
```

A.4 EDA and Visualization

```
promo = df.groupby("Promotion_Flag")["Units_Sold"].mean()
monthly = df.groupby(df["Date"].dt.to_period("M"))["Units_Sold"].mean()
corr = df[numeric_columns].corr()
```

A.5 Model Training and Evaluation

```
model = HistGradientBoostingRegressor(
    max_iter=300, learning_rate=0.05,
    max_leaf_nodes=31, l2_regularization=1.0,
    random_state=42
)
model.fit(train[features], train["Units_Sold"])
pred = model.predict(test[features])

mae = mean_absolute_error(test["Units_Sold"], pred)
rmse = mean_squared_error(test["Units_Sold"], pred) ** 0.5
r2 = r2_score(test["Units_Sold"], pred)
```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

The following supporting visualisations are included for completeness. The main report narrative uses the figures that directly answer the research questions; additional diagnostics are retained here.

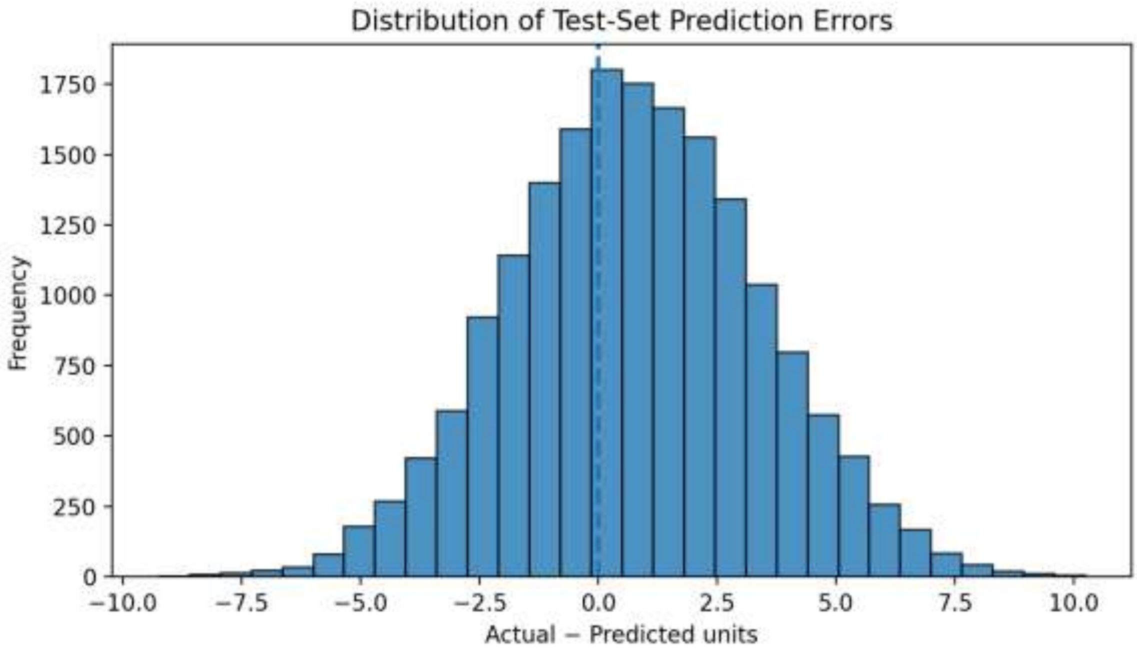

Figure 9. Test-Set Prediction Error Distribution (Supplementary View)

Supplier	Avg Units Sold	Avg Lead Time (days)	Avg Order Quantity
SUP_2	20.13	8.42	19.46
SUP_8	20.10	8.50	19.18
SUP_6	20.09	7.90	19.36
SUP_9	20.08	8.19	19.19
SUP_4	20.07	8.64	19.35
SUP_7	20.05	7.56	19.28
SUP_5	20.03	6.96	19.34
SUP_10	20.01	7.41	19.23
SUP_3	20.01	8.26	19.26
SUP_1	19.99	8.11	19.09

Table 7. Supplier-Level Supporting Summary

Month	Avg Demand	Avg Forecast	Avg Inventory	Avg Reorder Point	Below Reorder
2024-01	22.80	22.82	527.47	300.07	3.99%
2024-02	27.20	27.21	459.95	300.07	7.52%
2024-03	29.88	29.90	459.07	300.07	8.05%
2024-04	29.84	29.80	454.42	300.07	8.28%
2024-05	27.21	27.16	459.08	300.07	7.47%
2024-06	22.67	22.69	460.49	300.07	6.39%
2024-07	17.41	17.44	466.50	300.07	4.71%
2024-08	12.78	12.83	477.49	300.07	3.23%
2024-09	10.17	10.31	472.32	300.07	2.61%
2024-10	10.27	10.31	475.72	300.07	2.66%
2024-11	13.08	13.17	474.30	300.07	3.56%
2024-12	17.59	17.58	469.87	300.07	4.61%

Table 8. Monthly Supporting Summary